

P2 LANGUAGES

2.1

a) SELECT FirstName, LastName
FROM customers
WHERE Country = "Brazil"

SQL

$\Pi_{\langle \text{FirstName}, \text{LastName} \rangle} (\sigma_{\langle \text{Country} = \text{"Brazil"} \rangle} (\text{customers}))$

AR

$\{t/\exists c (c \in \text{customers} \wedge c.\text{Country} = \text{"Brazil"})\}$

CRT

~~b)~~

SELECT c.FirstName, i.InvoiceDate, i.InvoiceID
FROM customers c
JOIN invoice i ON c.customerId = i.customerId

$\Pi_{\langle \text{FN}, \text{IDate}, \text{IID} \rangle} (\sigma_{\langle \text{customer} \bowtie \text{invoice} \wedge c.\text{customerId} = i.\text{customerId} \rangle})$

$\{t/\exists c, i (c \in \text{customers} \wedge i \in \text{invoice} \wedge c.\text{customerId} = i.\text{customerId})\}$

c)

SELECT ar.Name
FROM track t

JOIN album al ON t.albumId = al.albumId

JOIN artist ar ON al.artistId = ar.artistId

$\Pi_{\langle \text{ar.Name} \rangle} ((\text{track} \bowtie_{\langle t.\text{albumId} = \text{al.albumId} \rangle} \text{album}) \bowtie_{\langle \text{al.artistId} = \text{ar.artistId} \rangle} \text{artist})$

$\{t/\exists tr, ar, al (tr \in \text{track} \wedge ar \in \text{artist} \wedge al \in \text{album} \wedge tr.\text{albumId} = al.\text{albumId} \wedge ar.\text{artistId} = al.\text{artistId})\}$

d)

$\Pi_{\langle pl.Name \rangle} \left(\sigma_{\langle t.mediatype = "MP3G audio file" \rangle} \left(\left(\text{playlist} \bowtie_{\langle pl.playlistid = plt.playlistid \rangle} \text{playlisttrack} \right) \bowtie_{\langle plt.trackid = t.trackid \rangle} \text{track} \right) \bowtie_{\langle t.mediatypeid = mt.mediatypeid \rangle} \text{mediatype} \right)$

$\{ \tau / \exists tr, pl, plt, mt (\tau \in \text{track} \wedge pl \in \text{Playlist} \wedge plt \in \text{playlisttrack} \wedge mt \in \text{MediaType} \wedge pl.playlistid = plt.playlistid \wedge plt.trackid = tr.trackid \wedge tr.mediatypeid = mt.mediatypeid \wedge mt.Name = "MP3G audio file") \}$

e)

SELECT pl.Name
 FROM playlist pl
 JOIN track playlisttrack plt ON pl.playlistid = plt.playlistid
 JOIN track t ON t.trackid = plt.trackid
 JOIN album al ON t.albumid = al.albumid
~~WHERE~~ JOIN artist ar ON al.artistid = al.artistid
 WHERE ar.Name = "IRON MAIDEN"
 GROUP BY pl.playlistid, pl.Name
 HAVING COUNT(*) > 10.

f)

SELECT pl.Name, COUNT(DISTINCT al.albumid) AS CANTIDAD
 FROM playlist pl, playlisttrack
 LEFT JOIN plt ON plt.playlistid = pl.playlistid
 LEFT JOIN track t ON t.trackid = plt.trackid
 GROUP BY pl.Name

USO LEFT PARA CONTAR LOS 0S DE LAS VACÍAS

g) SELECT e.FirstName

FROM employee e

JOIN customer c ON e.employeeId = c.SupportRepId

JOIN invoice i ON i.customerId = c.customerId

JOIN invoice_line il ON i.invoiceId = il.invoiceId

WHERE (STRftime('%Y', 'now') - STRftime('%Y', e.BirthDate))

AND il.QUANTITY > 10.

h) SELECT c.FirstName, c.InvoiceId, c.InvoiceDate

FROM customer c

LEFT JOIN invoice i ON c.customerId = i.customerId

i) SELECT e.name

FROM employee e

*

WHERE il.quantity > 10

j) SELECT e.FirstName, e.LastName, b.FirstName, b.LastName

FROM employee e

JOIN employee b ON e.employeeId = b.ReportsTo

B

k) SELECT A

FROM employee e

LEFT JOIN employee b ON B

1) SELECT c.FirstName, AVG(count) AS promedio
 FROM customer c
 JOIN invoice i ON c.customerId = i.invoiceId
 JOIN invoiceLine il ON i.invoiceId = il.invoiceId
~~JOIN track t ON t.trackId = il.trackId~~
 GROUP BY c.FirstName, c.customerId, ~~count~~ SUM(il.quantity)

2m) SELECT totalAmount
 FROM employee e
 JOIN customer c ON e.employeeId = c.SupportRepId
 JOIN invoice i ON c.customerId = i.customerId
 JOIN invoiceLine il ON i.invoiceId = il.invoiceId
 JOIN track t ON t.trackId = il.trackId
 (GROUP BY e.employeeId, COUNT(*))
 JOIN genre g ON g.genreId = t.genreId
 WHERE g.name = 'Rock'

2.2

a) $\Pi_{\langle \text{BAR} \rangle} ((\text{SERVE} \bowtie_{\langle \text{S.BAR} = \text{F.BAR} \rangle} \text{FRECUENTA}) \bowtie_{\langle \text{F.PERSONA} = \text{G.PERSONA} \rangle} \text{GUSTA})$
 $\sigma_{\langle \text{PERSONA} = \text{PERSONA} = \text{"SUAN R."} \rangle}$
 $\{t/\exists s, g (s \in \text{SERVE} \wedge g \in \text{GUSTA} \wedge \text{S.CERVEZA} = \text{G.CERVEZA} \wedge \text{G.PERSONA} = \text{"SUAN R."})\}$

b) $\{t/\exists f, s, g (f \in \text{FRECUENTA} \wedge s \in \text{SERVE} \wedge g \in \text{GUSTA} \wedge f.\text{BAR} = s.\text{BAR} \wedge s.\text{CERVEZA} = g.\text{CERVEZA})\}$

$\Pi_{\langle \text{G.PERSONA} \rangle} ((\text{FRECUENTA} \bowtie \text{BAR}) \bowtie \text{SERVE})$

$$(i, j) \rightarrow (m, n) = |i - m| + |j - n|$$

$$g) \pi_{PERSONA}(GOSTA) - \pi_{PERSONA}((SIRVE \bowtie GOSTA) \bowtie FRECUENTA)$$

$$\{t / \forall s, g (s \in Sirve \wedge g \in Gosta \wedge s.cerveza = g.cerveza \Rightarrow \exists f (f \in Frecuenta \wedge s.Bar = f.Bar))\}$$

$$d) \{t / \forall s, g (s \in Sirve \wedge g \in Gosta \wedge s.cerveza = g.cerveza \Rightarrow \exists f (f \in Frecuenta \wedge s.Bar = f.Bar))\}$$

$$\pi_{PERSONA}((SIRVE \bowtie GOSTA) \div FRECUENTA)$$

2.3

a) SELECT al.Title

FROM album al

JOIN track t ON t.albumId = al.albumId

JOIN playlisttrack plt ON t.trackId = plt.trackId

JOIN playlist pl ON pl.playlistId = plt.playlistId

GROUP BY al.albumId, al.~~album~~Title

HAVING

COUNT(DISTINCT pl.playlistId) = SELECT COUNT(*) FROM playlist

* Según el CHAT GPT no se puede hacer sin el SELECT adentro.

⑤ SELECT ar.Name, COUNT(DISTINCT al.albumId)
 FROM artist ar
 JOIN album al ON ar.artistId = al.artistId
 JOIN track t ON t.~~albumId~~ = al.albumId
 JOIN playlisttrack plt ON plt.trackId = t.trackId
 JOIN playlist pl ON pl.playlistId = plt.playlistId
 GROUP BY ar.Name, ar.artistId

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2.4
 a) SELECT pl.Name
 FROM playlist pl
 JOIN playlisttrack plt ON pl.playlistId = plt.playlistId
 JOIN track t ON plt.trackId = t.trackId
 JOIN album al ON al.albumId = t.albumId
 JOIN artist ar ON ar.artistId = al.artistId
 WHERE ar.Name IS NOT "Black Sabbath" AND
 ar.Name IS NOT "Chico Buarque"

{t/∃pl, plt, t, al, ar (pl ∈ Playlist ∧ plt ∈ playlisttrack ∧ 6... 6...
 ∧ pl.playlistId = plt.playlistId
 :
 ∧ ar.Name ≠ "Black Sabbath" ∧
 ∧ ar.Name ≠ "Chico Buarque") }

Tipename (Playlist $\bowtie$ PlaylistTrack $\bowtie$ Track $\bowtie$ Album $\bowtie$ Artist)

- Tipename (Artist $\in$ "Black Sabbath" $\wedge$ name = "Chicago") (A)

b) SELECT c.FirstNome, c.LastNome
FROM customer c

JOIN invoice i ON i.customerId = c.customerId

JOIN invoiceitem il ON il.invoiceId = i.invoiceId

JOIN track t ON t.trackId = il.trackId

JOIN genre g ON g.genreId = t.genreId

GROUP BY c.customerId, c.FirstNome, c.LastNome

HAVING COUNT(DISTINCT g.genreId) = 1

$\{t/\exists c, i, il, e, g (c \in \text{Customer} \wedge \dots \wedge$

$\Rightarrow \{t/\exists g' (g' \in \text{Genre}$
 $\wedge g'.\text{genreId} = t.\text{genreId} \wedge$
 $g' \neq g)\}$

(2.5)

a) SELECT a.nombreActor
FROM actor

~~JOIN serie s ON s.actorId = a.actorId~~

JOIN participacion p ON p.actorId = a.actorId

JOIN serie s ON p.serieId = s.serieId

WHERE s.nombreSerie = "Friends" AND a.edad > 30

b) SELECT c.member Canal

FROM canal c

JOIN translate t ON c.translate = t.id Canal

JOIN serie s ON s.idSerie = t.idSerie

JOIN genera g ON g.idGenera = s.idGenera

WHERE g.member Genera = "comedra"

c) a) ...

WHERE s.memberSerie = "Friends" ^{AND} s.idSerie > 30 AND
s.member > 2000

d) c) per con < 2000

e) SELECT s.idSerie

FROM serie s

ORDER BY member

LIMIT 10

f) SELECT a.idActor

FROM actor a

JOIN participa p ON p.idActor = a.idActor

JOIN serie s ON s.idSerie = p.idSerie

GROUP BY a.idActor

HAVING COUNT(DISTINCT idSerie) > 2